

GeoFlux

Exploring global commodity trade, 1988 – 2016

Team Jamie · Jason · Coco · Lucas

Dataset UN Global Commodity Trade Statistics + Kaggle International Trade DB

Stack Nuxt 3 · Mapbox GL · D3.js · Tailwind CSS

01 — OVERVIEW

What is GeoFlux?

GeoFlux is an interactive web application for exploring global commodity trade from 1988 to 2016. The app is built around a full-screen 3D globe (using Mapbox), and the main idea was to let users navigate trade data the same way they might use Google Earth — by spinning the globe, clicking on countries, or searching for whatever they want to look at.

We used the United Nations Global Commodity Trade Statistics dataset along with an international trade database from Kaggle. Our main goal was to take a fairly large and complicated dataset and make it understandable, without throwing too much information at the user at once.

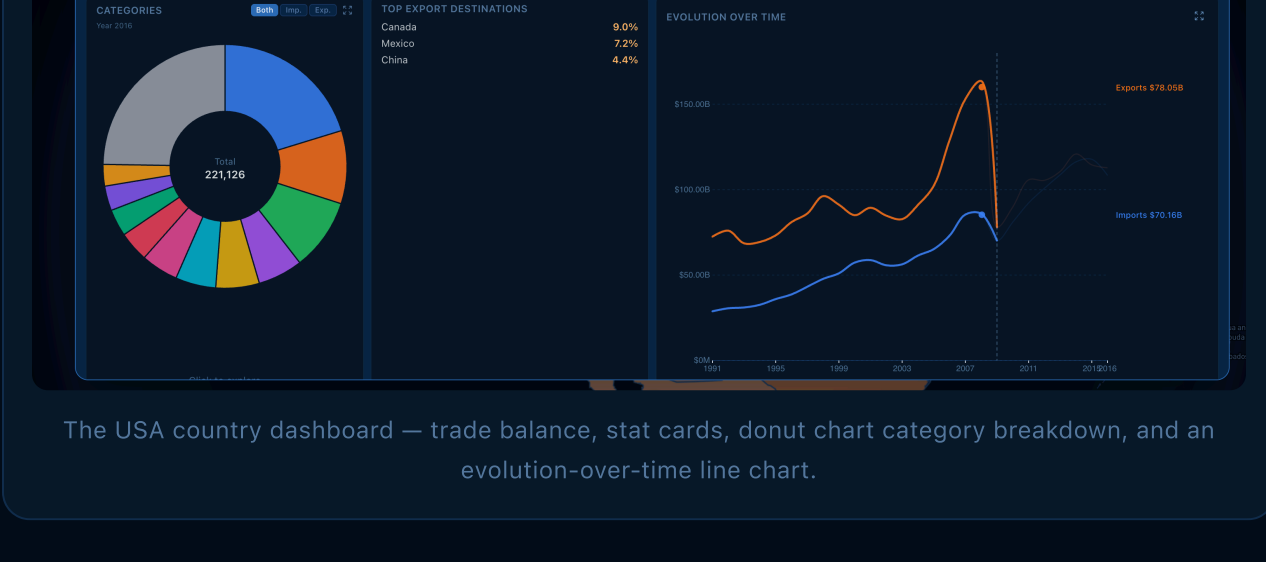


The globe works as both the navigation interface and a visualization. Countries can be colored based on import or export volume. There is also a toggle that draws lines between countries and their biggest trading partners, which turns the raw numbers into something you can actually see.

● Export destinations ● Import sources ● Both (import + export)

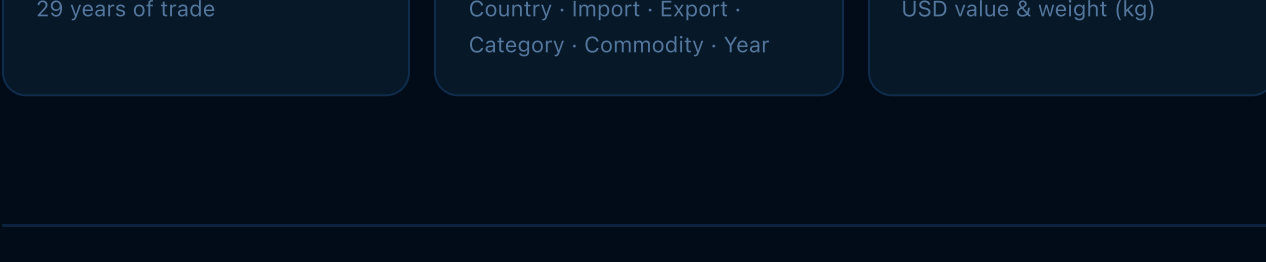
The user can also switch the metric between trade **value (USD)** and **weight (kg)**, which is useful because a country's value footprint and its weight footprint are often quite different.

When the user clicks on a country (or picks something from the search bar), a detail page slides over the globe. These dashboards are the main part of the app. The country page shows the trade balance, total imports and exports, a category breakdown as a donut chart, and a line chart showing how trade changed over time.



From the country view users can go deeper into separate pages for import and export breakdowns, commodity categories, and individual commodities. Each of these has its own top countries, historical data, and stats like growth since 1988 and share of world trade.

There is also a year page that shows totals and top countries/categories for a given year, along with a short note about what was happening globally that year.



02 — DESIGN CHOICES

How it was built

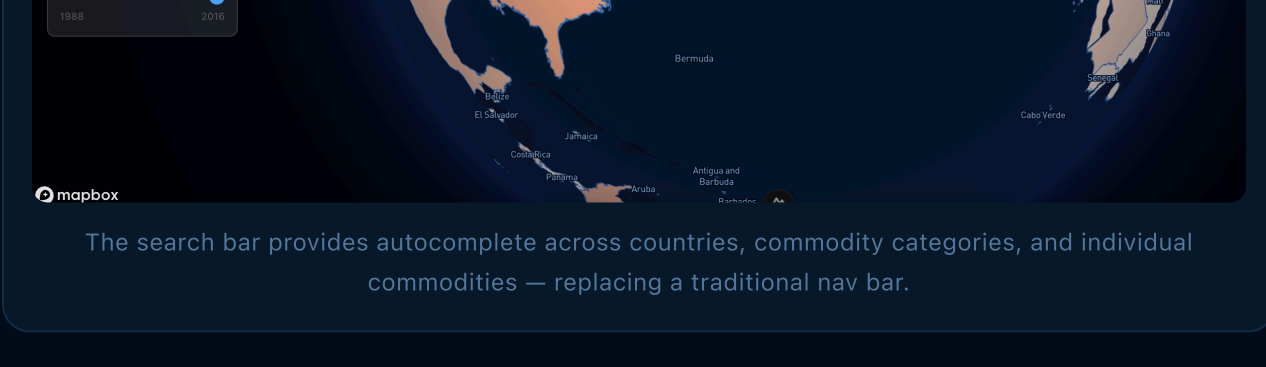
We focused a lot on **modularity**. Early on we built a set of base components: temporal bar charts, interactive pie charts, line charts, stat cards, and clickable country selectors. We reused these components across the different pages.

This made development faster since we weren't rewriting the same thing for each dashboard, but more importantly it kept the visual appearance **consistent** throughout the app. Because every page is built from the same pieces, the whole thing feels cohesive instead of like a bunch of separate views stitched together.

Temporal Bar Charts Interactive Pie Charts Line Charts Stat Cards Country Selectors

Shared across all dashboards

Search over nav bar. We considered adding a navigation bar, but we ended up with too many dashboards for that to scale cleanly. A nav bar listing all of them would have been cluttered. The search bar handled the same problem more cleanly — users can jump directly to any country, category, commodity, or year.



We also wanted the app to be more than just dashboards full of numbers, so we added visualizations directly on the map. The main one is the trade lines, which draw connections between a country and its primary import and export partners so you can see those relationships visually.

We also added a view that highlights the countries with the highest imports and exports across the globe. A lot of this information shows up numerically in the dashboards too, but we wanted to give users a more visual way to understand it rather than making them read everything off a table.

03 — CHALLENGES

What we worked through

Merging Two Datasets
The biggest challenge was merging the two datasets we used. Since they came from different sources, the figures didn't line up exactly — the same country in the same year could have noticeably different trade values depending on which dataset we pulled from. Rather than trying to force the numbers to match, we decided to work with **percentages and relative comparisons** where the two didn't agree. This let us show how countries compared to each other in terms of trade without claiming a precise dollar figure we couldn't actually back up across both sources.

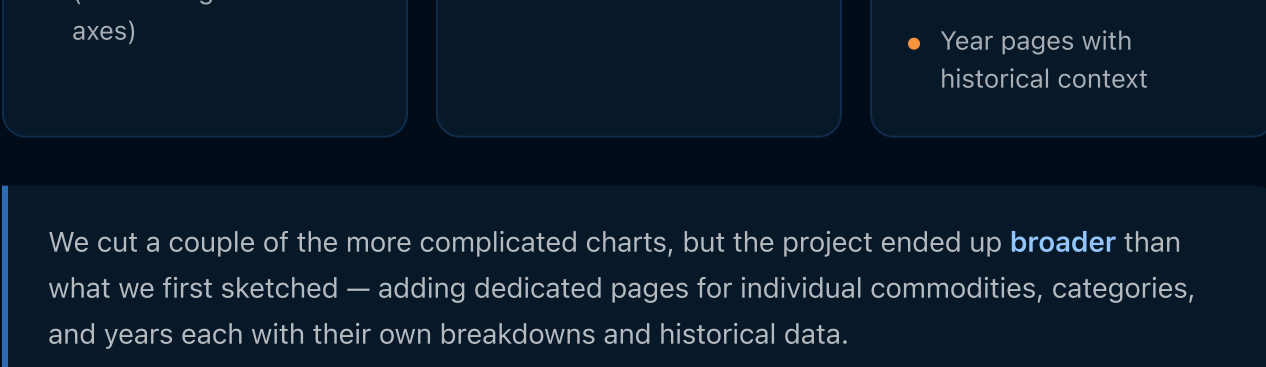
Missing Data
Both datasets had gaps — missing columns in some places and entire countries absent in others. Because of this we had to drop a number of countries from the final dataset and set them to be **non-interactable on the globe**. Users can still see them, but can't click through to a dashboard that would just be empty. This is a real limitation of our data coverage and something we'd want to address if we had more complete sources.

Performance
When we tried loading the raw data directly, everything was extremely slow — the page took a long time to load anything because of how much data there was to process on the fly. To fix this, we **preprocessed all of the data ahead of time into structured JSON files**. This meant the heavy work happened once, before deployment, instead of every time a user opened a page, and the app could then pull from these pre-built files and render the visualizations quickly.

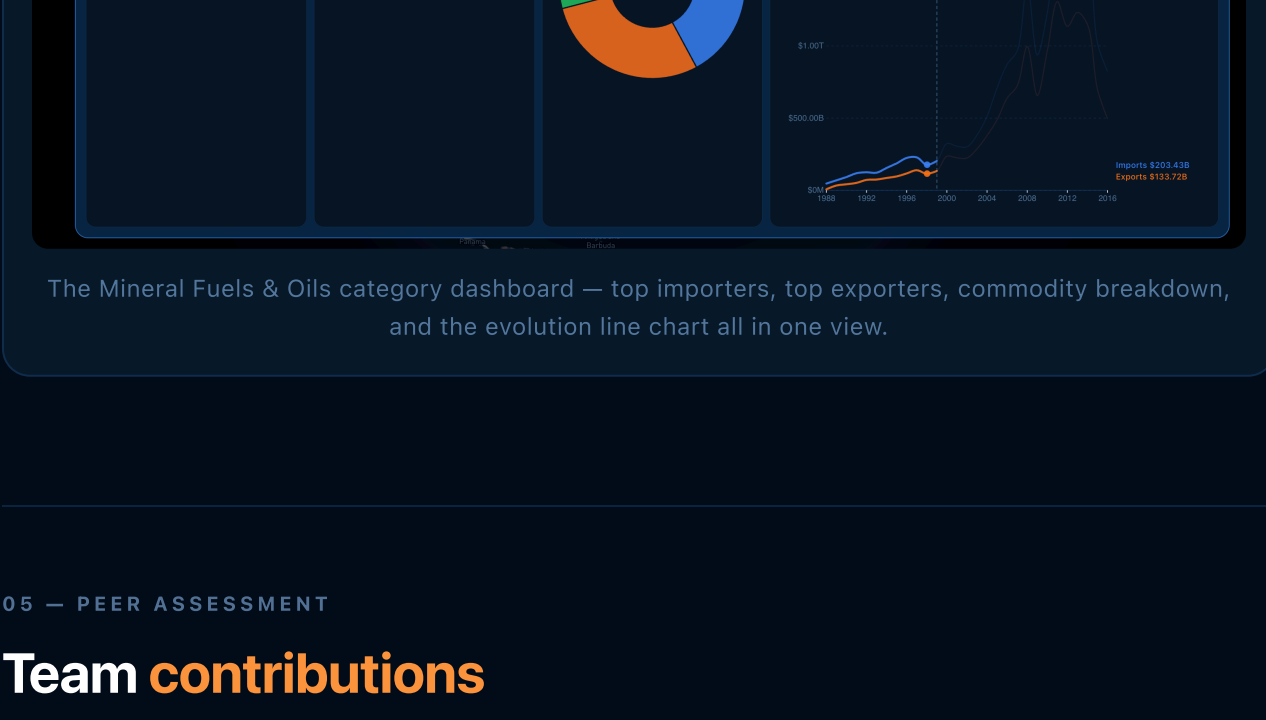
04 — CHANGES FROM ORIGINAL DESIGN

What changed

We dropped a few visualizations from our original design because they were too hard to understand. Both tried to pack too much into one chart, and instead of being informative they just ended up confusing.



We cut a couple of the more complicated charts, but the project ended up **broadier** than what we first sketched — adding dedicated pages for individual commodities, categories, and years each with their own breakdowns and historical data.



05 — PEER ASSESSMENT

Team contributions

Jamie **FRONTEND**
Expanded the country highlighting feature so countries can be highlighted based on USD, weight, and individual categories, broken down by imports or exports. Collaborated on setting up the trade connection lines between countries, and contributed to the process book.

Jason **SEARCH & DOCS**
Set up the search functionality that lets users navigate directly to any dashboard. Collaborated on the trade connection lines between countries, documented the project's design and architecture, and contributed to the process book.

Coco **CORE & D3**
Set up Mapbox and the core dashboard design, and wrote the majority of the D3 components used throughout the dashboards. Added the initial Vue components for search, trade connections, and country highlighting, which the rest of the team built on top of.

Lucas **MAP & CHARTS**
Got the global map coloring working and disabled hover and click interactions for countries with no data. Implemented the pie chart tooltips and a reusable bar chart D3 component used for global stats, imports, exports, countries, commodities, and categories.